Design and Documentation of a College Department Website Using HTML

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1. Introduction

This report explains how a simple, multi-section website for the **Department of Computer Science** of FISAT Engineering College was designed using only HTML5 (with a little CSS for presentation). The site has six sections: About, Programmes, Faculty, Facilities, Enquiry and Contact. All code is contained in a single file, `index.html`, which opens in any web browser. The college is FISAT Engineering College, Angamaly, Kerala; the faculty names and some contact details are sample data to be replaced with real ones.

2. HTML Document Structure

Every HTML page follows a standard skeleton. **<!DOCTYPE html>** tells the browser the page uses HTML5, so it renders in standards mode. **<html>** is the root element that wraps all content; the *lang* attribute helps screen readers and search engines. **<head>** holds information about the page that is not displayed (character set, viewport, styles). **<title>** sets the text on the browser tab and in search results. **<body>** contains everything visible to the user.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Department of Computer Science | FISAT Engineering College</title>
</head>
<body>
  ... visible page content ...
</body>
</html>
```

3. Headings and Paragraphs

Headings **<h1>** to **<h6>** create a hierarchy of importance; the page uses one h1 for the department name, h2 for each section and h3 for sub-topics. Paragraphs use **<p>**. Text formatting tags used: **** (important text), **** (emphasis), **
** (line break) and **<hr>** (horizontal rule).

```
<h2>About the Department</h2>
<p>The Department of Computer Science was established in <strong>2005</strong>.
It offers <em>industry-oriented</em> education in programming, data and networks.</p>
<hr>
```



Figure 1: Header with logo, main heading and navigation

4. Lists

An **unordered list** (<ul>) shows items with bullets, and an **ordered list** (<ol>) numbers them. Each item is written in an <li> element. The department's five subjects are shown below.

```
<ul>
  <li>Programming in C and C++</li>
  <li>Data Structures and Algorithms</li>
```

```
  <li>Database Management Systems</li>
  <li>Web Technology</li>
  <li>Operating Systems</li>
</ul>
```

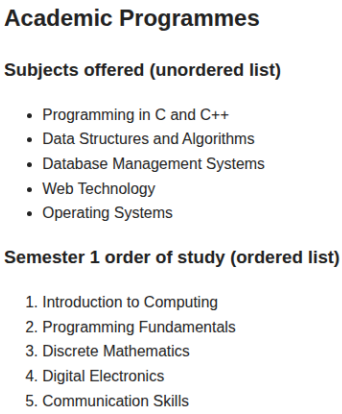


Figure 2: Unordered and ordered lists

5. Hyperlinks

The anchor tag **<a>** with the *href* attribute creates a link. External links use a full URL; *target="_blank"* opens a new tab and *rel="noopener"* adds security. Internal links such as `href="#faculty"` jump to the element with the matching *id*, which is how the navigation bar works. `mailto:` links open the email client.

```
<a href="https://www.fisat.ac.in" target="_blank" rel="noopener">college website</a>
<a href="#faculty">Faculty</a>
<a href="mailto:info@fisat.ac.in">info@fisat.ac.in</a>
```

6. Images

The **<img>** tag is a self-closing element that embeds an image. *src* gives the file path, *alt* gives alternative text for accessibility and for when the image fails to load, and *width* sets the size. The department logo is embedded in the page (the snippet shows the file path form).

```
<img src="images/logo.svg" alt="Department of Computer Science logo" width="80">
```

7. Tables

Tables display data in rows and columns. **<table>** is the container, **<thead>**/**<tbody>** group header and body rows, **<tr>** is a row, **<th>** a header cell and **<td>** a data cell.

```
<table>
  <thead>
    <tr><th>Name</th><th>Designation</th><th>Area of Specialization</th></tr>
  </thead>
  <tbody>
    <tr><td>Dr. A. Nair</td><td>Professor and Head</td><td>Machine Learning</td></tr>
    <tr><td>Dr. B. Menon</td><td>Associate Professor</td><td>Computer Networks</td></tr>
    <tr><td>Ms. C. Thomas</td><td>Assistant Professor</td><td>Database Systems</td></tr>
    <tr><td>Mr. D. Kumar</td><td>Assistant Professor</td><td>Web Technologies</td></tr>
    <tr><td>Ms. E. Joseph</td><td>Assistant Professor</td><td>Cyber Security</td></tr>
  </tbody>
</table>
```

Our Faculty

Faculty members

Name	Designation	Area of Specialization
Dr. A. Nair	Professor and Head	Machine Learning
Dr. B. Menon	Associate Professor	Computer Networks
Ms. C. Thomas	Assistant Professor	Database Systems
Mr. D. Kumar	Assistant Professor	Web Technologies
Ms. E. Joseph	Assistant Professor	Cyber Security

Figure 3: Faculty table

8. Forms

Forms collect user input. The **<form>** element defines where (*action*) and how (*method*) data is sent. **<input>** has several types (text, email, tel); *required* and *pattern* provide built-in validation. **<label for>** is linked to an input's *id* for accessibility, **<select>** gives a drop-down, and **<button type="submit">** sends the form.

```
<form action="#" method="post">
  <label for="name">Name</label>
  <input type="text" id="name" name="name" required>
  <label for="email">Email</label>
  <input type="email" id="email" name="email" required>
  <label for="phone">Phone</label>
  <input type="tel" id="phone" name="phone" pattern="[0-9]{10}" required>
  <label for="course">Course Preference</label>
  <select id="course" name="course">
    <option>BSc Computer Science</option>
    <option>BCA</option>
    <option>MSc Computer Science</option>
  </select>
  <button type="submit">Submit</button>
</form>
```

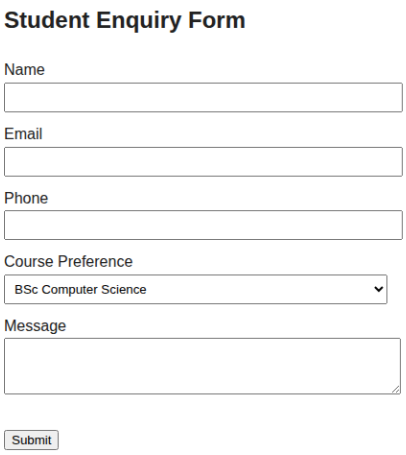


Figure 4: Student enquiry form

9. Semantic HTML

Semantic elements describe the *meaning* of content rather than just its appearance. This improves accessibility, SEO and code readability. Elements used in this site:

Element	Purpose in this website
<header>	Top banner with logo, department name and college name
<nav>	Navigation links to each section
<main>	Main unique content of the page
<section>	Groups related content (About, Programmes, Faculty ...)
<article>	Self-contained items such as each facility
<footer>	Copyright information at the bottom

```
<header> ... </header>
<nav> <a href="#about">About</a> ... </nav>
<main>
  <section id="faculty"> ... </section>
  <section id="facilities"> <article> ... </article> </section>
</main>
<footer> &copy; 2026 Department of Computer Science </footer>
```

10. Conclusion

The project shows how core HTML elements combine to build a clear, accessible multi-section website. Semantic structure, correctly labelled forms, descriptive links and alt text make the site easy for users and search engines to understand. The site can be extended with more pages, external CSS and JavaScript validation.

References

1. MDN Web Docs, "HTML: HyperText Markup Language", developer.mozilla.org/en-US/docs/Web/HTML
2. W3Schools, "HTML Tutorial", w3schools.com/html
3. WHATWG, "HTML Living Standard", html.spec.whatwg.org
4. web.dev, "Learn HTML", web.dev/learn/html